

Time Series Practice Questions

1. You are given the following time-series for monthly building settlement (mm):
5, 7, 10, 12, 15, 14, 12, 9, 7, 6, 7, 9
Identify whether each of these components exist in this time series: trend, seasonality and noise. Approximately sketch the decomposition.

Answer: Trend: weak

Seasonality: Yes (rising till 5th month) and then falls

Noise: Yes

2. You are working on modelling displacement of bridge girder. You have collected the following data:

$$t: 0, 1, 2, 3, 4$$

$$x(t): 10, 13, 15, 18, 20$$

Assuming linear trend model $x(t) = a + bt + \epsilon_t$, find the Least-Square Estimate for a and b. What type of error are we implicitly assuming here?

Answer: Design matrix, $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}$, $x = \begin{bmatrix} a \\ b \end{bmatrix}$ and $y = \begin{bmatrix} 10 \\ 13 \\ 15 \\ 18 \\ 20 \end{bmatrix}$

$$\hat{a} = 10.2 \text{ and } \hat{b} = 2.5$$

Since we are using LS as best estimator, we are assuming that the noise is Gaussian White Noise.

3. Given the following observations:

$$y = Ax + \epsilon$$

$$y = \begin{bmatrix} 5 \\ 8 \\ 6 \end{bmatrix}, A = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \text{cov}(\epsilon) = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix}$$

Compute the BLUE of x.

Answer:

$$\hat{x}_{BLUE} = (A^T \Sigma^{-1} A)^{-1} A^T \Sigma^{-1} y = 6.13$$

4. In question 3, compute the LS estimate. How do the two compare?

Answer:

$$\hat{x}_{LS} = (A^T A)^{-1} A^T y = 6.33$$

5. You observe the following three time series:

$$1) x_t = 5 + \epsilon_t$$

$$2) x_t = 2t + \epsilon_t$$

$$3) x_t = 0.7x_{t-1} + \epsilon_t$$

Assuming ϵ_t is white noise, determine which of these are stationary.

Answer:

- 1) Mean = 5, Std = σ_{noise} . Does not change with time, stationary
- 2) Mean = $2t$, Std = σ_{noise} . Mean changes with time, so non-stationary
- 3) Mean = $mean(x_0)$, Std = σ_{noise} , $Cov(x_t, x_{t+\tau}) = 0.7^\tau$. Does not change with time, stationary. AR(1) process

6. For the process

$$x_t = 3 + \epsilon_t$$

Where ϵ_t is white noise with standard deviation σ^2 , find the mean of x_t , μ and covariance $Cov(x_t, x_{t+\tau}) = \mathbb{E}((x_t - \mu)(x_{t+\tau} - \mu))$.

Answer:

$$\begin{aligned}\mu &= \mathbb{E}[x_t] = \mathbb{E}[3 + \epsilon_t] = \mathbb{E}[3] + \mathbb{E}[\epsilon_t] = 3 \\ cov(x_t, x_{t+\tau}) &= \mathbb{E}((3 + \epsilon_t - 3)(3 + \epsilon_{t+\tau} - 3)) = \\ \mathbb{E}(\epsilon_t \epsilon_{t+\tau}) &= \begin{cases} \sigma^2 & \text{if } \tau = 0 \\ 0 & \text{if } \tau \neq 0 \text{ (Uncorrelated since white noise)} \end{cases}\end{aligned}$$

7. For the AR(1) model:

$$S_t = 0.6S_{t-1} + \epsilon_t$$

Where ϵ_t is white noise with standard deviation 4. Find the

- a) Is this a stationary time series?
- b) Mean μ
- c) Variance of $S_t = \mathbb{E}((S_t - \mu)(S_t - \mu))$
- d) Lag 1 covariance, $cov(S_t, S_{t+1}) = \mathbb{E}((S_t - \mu)(S_{t+1} - \mu))$

Answer:

- a) Yes
- b) $\mu = \mathbb{E}[S_t] = \mathbb{E}[0.6S_{t-1} + \epsilon_t] = \mathbb{E}[\epsilon_t + 0.6\epsilon_{t-1} + 0.6^2\epsilon_{t-2} + \dots + 0.6^n\epsilon_0] = 0$
- c) $Var(S_t) = Var(0.6S_{t-1} + \epsilon_t) = 0.6^2Var(S_{t-1}) + \sigma_{noise}^2 = 0.36Var(S_t) + 16$
 $0.64Var(S_t) = 16 \Rightarrow Var(S_t) = 25$
- d) $cov(S_t, S_{t+1}) = \mathbb{E}((S_t - \mu)(S_{t+1} - \mu)) = \mathbb{E}((S_t)(S_{t+1})) = \mathbb{E}((S_t)(0.6S_t + \epsilon_t)) = \mathbb{E}((S_t)(0.6S_t + \epsilon_t)) = 0.6\mathbb{E}(S_t^2) + \mathbb{E}(\epsilon_t S_t) = 0.6 * Var(S_t) = 15$

8. For the given AR(1) model:

$$S_t = 0.8S_{t-1} + \epsilon_t$$

$$S_{10} = 20$$

We know ϵ_t is white noise.

What is the mean of S_{11} , S_{12} and S_{13} ?

Answer:

$$\begin{aligned}\mu_{S_{11}} &= \mathbb{E}[0.8 * S_{10} + \epsilon_t | S_{10}] = \mathbb{E}[0.8 * S_{10} | S_{10}] + \mathbb{E}[\mu_t] = 16 + 0 = 16 \\ \mu_{S_{12}} &= \mathbb{E}[0.8 * S_{11} + \epsilon_t | S_{10}] = \mathbb{E}[0.8 * S_{11} | S_{10}] + \mathbb{E}[\mu_t] = 16 * 0.8 + 0 \\ &= 12.8 \\ \mu_{S_{13}} &= \mathbb{E}[0.8 * S_{12} + \epsilon_t | S_{10}] = \mathbb{E}[0.8 * S_{11} | S_{10}] + \mathbb{E}[\mu_t] = 12.8 * 0.8 + 0 \\ &= 10.24\end{aligned}$$

9. Using the AR(1) model in Question 8, derive 1-step and 2-step covariances if $\sigma^2 = 9$.

Answer:

$$Var(S_t) = \frac{\sigma^2}{1 - \phi^2} = \frac{9}{1 - 0.64} = 25$$

1-step covariance:

$$Cov(S_t, S_{t+1}) = \phi Var(S_t) = 0.8 * 25 = 20$$

2-step covariance:

$$Cov(S_t, S_{t+2}) = \phi^2 Var(S_t) = 0.64 * 25 = 16$$